The Projective-Orbit Retry Theorem, Expanded

A detailed proof of Theorem 7.1 and a complete calculation of the unit-square distance matrix

Quantum Hodology Research Project

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Abstract

Theorem 7.1 of *Exact Low-Dimensional Hodology* gives a simple but powerful recipe: encode a group orbit in a quantum system, make a displacement of metric length r succeed with probability $1/r$, and charge one unit for every attempt. Retrying then costs r on average, and the triangle inequality proves that no adaptive sequence of other actions is cheaper. This note explains every assumption and every inequality in that argument. It also corrects a left-versus-right invariance convention that is harmless for the abelian examples in the paper but matters for nonabelian groups. Finally, it derives every entry of the four-by-four matrix D in Eq. (43), checks the Bellman equations directly, and performs the complete Schoenberg calculation showing that D is exactly the Euclidean metric of a unit square. Finally, it removes the assumption that the agent is handed its group label: a tetrahedral SIC outcome prepares and reports the current operational anchor, and SIC-outcome goals retain the same square geometry after subtraction of a derived common reporting overhead. The expanded control analysis makes the reward convention explicit, evaluates all four actions in every state, and solves the discounted problem in closed form. It shows that the square is specifically an undiscounted expected-hitting-time geometry: discounting can change the optimal policy and collapse its spatial distinctions.

Contents

1 What the theorem is trying to accomplish

Suppose an agent has operational situations indexed by elements x of a group G. An action indexed by $h \in G$ attempts to translate the situation. The desired difficulty of displacement h is a prescribed group metric length

$$r_h = d_G(\mathsf{e}, h).$$

The construction turns this length into a waiting time. One attempt costs one intervention and succeeds with probability

$$p_h = \frac{1}{r_h}.$$

If failure changes nothing, repeated attempts have geometric mean waiting time $1/p_h = r_h$. This immediately supplies a policy whose expected cost equals the desired distance. The nontrivial half of the theorem is to prove that a clever combination of shorter displacements, stochastic choices, or history-dependent actions cannot do better.

The proof uses the metric distance to the goal as a *potential*:

$$V_g(x) = d_G(x, g).$$

The triangle inequality bounds how much any successful action can reduce this potential. The probability $p_h = 1/r_h$ is calibrated so that the *expected* decrease of V_g in one unit-cost intervention is at most one. Reducing an initial potential $V_g(x)$ to zero therefore costs at least $V_g(x)$ on average. Direct retry attains this bound.

2 The assumptions, one by one

2.1 A group of displacements

A group G has an identity e, inverses, and an associative product. The product records composition of displacements. For example, in the square construction

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2,$$

with componentwise addition modulo two. Its elements are $00, 10, 01, 11$. Every nonidentity element is its own inverse.

2.2 A translation-invariant metric

A metric d_G is *left invariant* if

$$d_G(ax, ay) = d_G(x, y) \qquad \text{for every } a, x, y \in G,$$

and *right invariant* if

$$d_G(xa, ya) = d_G(x, y).$$

It is *bi-invariant* if it has both properties. On an abelian group the left and right translations coincide, so either invariance implies the other.

The normalization

$$\min_{h \neq \mathsf{e}} d_G(\mathsf{e}, h) = 1$$

is not cosmetic. It guarantees $r_h \geq 1$, hence $0 < p_h = 1/r_h \leq 1$, as required for a probability. It also fixes the unit of cost: the shortest nontrivial displacement costs one expected intervention.

2.3 A projective quantum orbit

A projective unitary representation assigns a unitary U_g to every $g \in G$, with multiplication respected up to global phase:

$$U_g U_h = e^{i\phi(g,h)} U_{gh}.$$

Global phase disappears under conjugation, so the density operators

$$\rho_g = U_g \rho_0 U_g^\dagger$$

carry an exact group action. The distinct-orbit assumption is

$$\rho_g = \rho_h \quad \Longrightarrow \quad g = h.$$

It says that two different group labels do not denote exactly the same physical density operator. It does *not* say that the states are orthogonal or perfectly distinguishable in one shot.

The cost proof itself is a statement about the induced controlled group process. Assumption (2.3) supplies its physical quantum realization and prevents an exact stabilizer from silently identifying two purported places.

2.4 The retry instrument

For every nonidentity displacement h, set

$$K_{\mathrm{s}}^{(h)} = \sqrt{p_h}\,U_h, \qquad K_{\mathrm{f}}^{(h)} = \sqrt{1-p_h}\,I, \qquad p_h = \frac{1}{d_G(\mathsf{e},h)}. \tag{1}$$

The labels s and f mean success and failure. This is a valid two-outcome quantum instrument because

$$K_{\mathrm{s}}^{(h)\dagger}K_{\mathrm{s}}^{(h)} + K_{\mathrm{f}}^{(h)\dagger}K_{\mathrm{f}}^{(h)} = p_h U_h^\dagger U_h + (1-p_h)I \tag{2}$$

$$= p_h I + (1-p_h)I = I. \tag{3}$$

Conditional on failure, normalization removes the scalar and the state is unchanged. Conditional on success, normalization removes $\sqrt{p_h}$ and the state is conjugated by U_h.

3 A convention issue in the printed statement

The printed theorem lets success act by left multiplication, $x \mapsto hx$, but assumes only that d_G is left invariant. It then uses

$$d_G(x, hx) = d_G(\mathsf{e}, h).$$

For a general nonabelian group, that equality follows from *right* invariance, not left invariance: right multiplication of both arguments by x^{-1} sends (x, hx) to (e, h).

There are two equally valid repairs.

1. Keep the printed dynamics $x \mapsto hx$ and assume that d_G is right invariant. The direct displacement is $h = gx^{-1}$.
2. Keep a left-invariant metric but use right-action dynamics $x \mapsto xh$. The direct displacement is $h = x^{-1}g$.

The paper's exact examples use abelian groups such as $\mathbb{Z}_2^2$, $\mathbb{Z}^2$, and finite phase tori. Their metrics are bi-invariant, so the displayed proof and every reported result are unchanged. The distinction matters only when generalizing the theorem to nonabelian action groups.

For definiteness, the next section gives the corrected left-action/right- invariant form. Reversing every product gives the left-invariant/right-action form verbatim.

4 The theorem with a fully explicit proof

Theorem 4.1 (Projective-orbit retry construction, convention-correct form)**.** *Let G be finite or countable and let d_G be a right-invariant metric normalized by* (2.2)*. Suppose G has a projective unitary representation and a fiducial density operator satisfying* (2.3)*. Under action $h \neq \mathsf{e}$, failure leaves x fixed and success sends x to hx, with the instrument* (1)*. Every attempt costs one. If the goal is g, then the optimal expected hitting cost is*

$$D(x, g) = d_G(x, g).$$

Proof. We separate the proof into validity, achievability, and optimality.

1. The quantum actions are valid. Equation (3) proves trace preservation after summing over the two reported branches. Because $d_G(\mathsf{e}, h) \geq 1$, both coefficients are real and lie between zero and one.

2. Direct retry achieves the metric distance. Assume $x \neq g$ and choose

$$h_* = gx^{-1}.$$

On success, $h_* x = g$, so one successful event reaches the goal. On failure, the state and group coordinate remain at x, so the same experiment starts again from exactly the same situation. The number T of attempts until the first success therefore has

$$\Pr(T = t) = (1 - p_{h_*})^{t-1} p_{h_*}, \qquad t = 1, 2, \ldots.$$

This is the geometric distribution, whose mean is

$$\mathbb{E}T = \frac{1}{p_{h_*}} = d_G(\mathsf{e}, h_*).$$

Right invariance, applied by multiplying both arguments by x, gives

$$d_G(\mathsf{e}, gx^{-1}) = d_G(x, g).$$

Thus direct retry has expected cost $d_G(x, g)$, proving

$$D(x, g) \leq d_G(x, g). \tag{4}$$

3. No single action can beat the distance potential. Define

$$V(y) = d_G(y, g).$$

If the current point is y and action h is selected, the expected one-step cost plus future potential is

$$Q_h(y) = 1 + (1 - p_h)V(y) + p_h V(hy). \tag{5}$$

The reverse triangle inequality follows from the ordinary triangle inequality:

$$V(y) = d_G(y, g) \leq d_G(y, hy) + d_G(hy, g) = d_G(y, hy) + V(hy).$$

Consequently

$$V(y) - V(hy) \leq d_G(y, hy).$$

Right invariance gives

$$d_G(y, hy) = d_G(\mathsf{e}, h) = \frac{1}{p_h},$$

and hence

$$p_h\big[V(y) - V(hy)\big] \leq 1. \tag{6}$$

Rearranging (5),

$$Q_h(y) = V(y) + 1 - p_h\big[V(y) - V(hy)\big] \geq V(y). \tag{7}$$

Thus every allowed action has Bellman cost at least the metric potential.

4. Why this also rules out adaptive policies and mixtures. Conditioned on the complete past, a policy may choose any action, possibly at random. Inequality (6) says that the conditional expected decrease of V during that intervention is at most one for every pure choice. Averaging preserves the inequality for a randomized choice. After n interventions, the expected total decrease is therefore at most n. For a policy with hitting time τ, apply this statement to $\tau \wedge n$ and let $n \to \infty$. At the goal, $V = 0$, so every proper policy of finite expected duration obeys

$$V(x) \leq \mathbb{E}\tau.$$

Policies with infinite expected duration cannot improve the optimum. Hence

$$D(x, g) \geq V(x) = d_G(x, g). \tag{8}$$

Combining (4) and (8) proves equality. $\square$

4.1 Where equality occurs

For the direct action $h_* = gx^{-1}$, success lands at g, so $V(h_* x) = 0$, while $1/p_{h_*} = V(x)$. Therefore

$$Q_{h_*}(x) = 1 + \left(1 - \frac{1}{V(x)}\right) V(x) = V(x).$$

All alternative actions satisfy $Q_h(x) \geq V(x)$; some may tie, but none can be smaller. This is the precise sense in which the triangle inequality is the optimality certificate.

5 Eq. (43): the unit-square example

5.1 The group and its desired metric

Take

$$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{00, 10, 01, 11\}.$$

Addition is componentwise modulo two. We associate these labels with the four Euclidean points

$$00 \leftrightarrow (0, 0), \quad 10 \leftrightarrow (1, 0), \quad 01 \leftrightarrow (0, 1), \quad 11 \leftrightarrow (1, 1). \tag{9}$$

Define the translation-invariant group metric by its displacement lengths:

$$d_G(\mathsf{e}, 10) = 1, \qquad d_G(\mathsf{e}, 01) = 1, \qquad d_G(\mathsf{e}, 11) = \sqrt{2}, \tag{10}$$

and

$$d_G(x, g) = d_G(\mathsf{e}, g - x).$$

Here subtraction equals addition because every element has order two. The only nontrivial triangle inequality is $\sqrt{2} \leq 1 + 1$; hence this is indeed a metric. Since G is abelian, it is both left and right invariant.

5.2 The qubit orbit

The physical system is a qubit with fiducial Bloch vector

$$\boldsymbol{r}_{00} = \frac{1}{\sqrt{3}}(1,1,1).$$

Conjugation by Pauli X and Z generates four states:

$$\boldsymbol{r}_{00} = (+,+,+)/\sqrt{3}, \qquad \boldsymbol{r}_{10} = (+,-,-)/\sqrt{3},$$
$$\boldsymbol{r}_{01} = (-,-,+)/\sqrt{3}, \qquad \boldsymbol{r}_{11} = (-,+,-)/\sqrt{3}.$$

They are distinct and form a tetrahedron on the Bloch sphere. Every distinct pair has overlap

$$\mathrm{Tr}(\rho_x \rho_g) = \frac{1}{3}.$$

Thus their *physical-state geometry* is equidistant. The square below is instead the geometry of optimal controlled cost.

5.3 Removing the hidden-location assumption with the SIC instrument

The labels $00, 10, 01, 11$ should not be supplied as an unobserved location register. The tetrahedral orbit itself defines a four-outcome qubit SIC. Let $\Pi_o = \rho_o$ be its rank-one projectors and define

$$M_o = \frac{1}{\sqrt{2}}\Pi_o, \qquad o \in \{00, 10, 01, 11\}. \tag{11}$$

Because a qubit SIC obeys $\sum_o \Pi_o = 2I$,

$$\sum_o M_o^\dagger M_o = \frac{1}{2}\sum_o \Pi_o = I.$$

Thus $\{M_o\}$ is a valid four-outcome instrument.

If its input is orbit state ρ_x, then

$$p(o \mid x) = \mathrm{Tr}(M_o \rho_x M_o^\dagger) = \frac{1}{2}\,\mathrm{Tr}(\Pi_o \Pi_x) = \begin{cases} 1/2, & o = x, \\ 1/6, & o \neq x. \end{cases} \tag{12}$$

The outcome does not reveal the *premeasurement* orbit label perfectly; four nonorthogonal qubit states cannot be perfectly discriminated. The key fact is instead the conditional update

$$\frac{M_o \rho_x M_o^\dagger}{p(o \mid x)} = \Pi_o. \tag{13}$$

Every possible outcome o prepares the corresponding tetrahedral state. Immediately after the SIC action, the agent therefore knows its current operational state: it is the last observed outcome token. There is no hidden coordinate inference in this statement. The binary strings are convenient theorist names and may be replaced by arbitrary colors or integers.

The agent can learn what opaque movement buttons do as well. Starting from an observed SIC anchor o, it presses a button and performs the same SIC again. The next-outcome law has its $1/2$ peak at the translated anchor and $1/6$ elsewhere. Repeating this common test recovers the induced permutations of the four outcome tokens, up to the unavoidable choice of origin, axes, and signs. Thus both "where I am" and "what this action does" can be grounded in observed statistics rather than privileged binary coordinates.

5.4 The three displacement actions

The axial displacements 10 and 01 have length one, so their retry probabilities are one. They are deterministic Pauli channels:

$$10 : \rho \mapsto X\rho X, \qquad 01 : \rho \mapsto Z\rho Z.$$

The diagonal displacement 11 is implemented by Y, projectively equivalent to XZ, and has

$$p_{11} = \frac{1}{\sqrt{2}}.$$

Therefore

$$K_{\mathrm{s}} = \sqrt{p_{11}}Y = 2^{-1/4}Y, \qquad K_{\mathrm{f}} = \sqrt{1 - \frac{1}{\sqrt{2}}}I. \tag{14}$$

Repeated diagonal attempts have expected cost

$$\frac{1}{p_{11}} = \sqrt{2}.$$

Taking two deterministic axial steps would cost 2, so direct diagonal retry is better. The theorem proves that no adaptive alternative costs less than $\sqrt{2}$.

5.5 Computing every entry of D

Order rows and columns as $00, 10, 01, 11$. The diagonal entries vanish because the agent is already at its goal:

$$D(00, 00) = D(10, 10) = D(01, 01) = D(11, 11) = 0.$$

Pairs differing in exactly one bit are axial neighbors and cost one:

$$\begin{aligned} D(00, 10) &= 1, & D(00, 01) &= 1, \\ D(10, 11) &= 1, & D(01, 11) &= 1. \end{aligned}$$

Pairs differing in both bits require displacement 11 and cost $\sqrt{2}$:

$$D(00, 11) = \sqrt{2}, \qquad D(10, 01) = \sqrt{2}.$$

The metric is symmetric, so reversing any pair gives the same entry. Hence

$$\boxed{D = \begin{pmatrix} 0 & 1 & 1 & \sqrt{2} \\ 1 & 0 & \sqrt{2} & 1 \\ 1 & \sqrt{2} & 0 & 1 \\ \sqrt{2} & 1 & 1 & 0 \end{pmatrix}}. \tag{15}$$

The same calculation can be displayed as a displacement table:

source	goal	group difference	optimal expected cost
00	10	10	1
00	01	01	1
00	11	11	$\sqrt{2}$
10	01	11	$\sqrt{2}$
10	11	01	1
01	11	10	1

6 A direct Bellman check for the square

It is useful to verify one column of D without invoking the theorem as a black box. Choose goal $g = 11$. From (15), the proposed values are

$$V(00) = \sqrt{2}, \qquad V(10) = 1, \qquad V(01) = 1, \qquad V(11) = 0.$$

At 00, deterministic X or Z gives

$$Q_X(00) = 1 + V(10) = 2, \qquad Q_Z(00) = 1 + V(01) = 2.$$

The diagonal action succeeds at 11 with probability $1/\sqrt{2}$ and otherwise remains at 00, so

$$\begin{aligned} Q_Y(00) &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) V(00) + \frac{1}{\sqrt{2}} V(11) \\ &= 1 + \left(1 - \frac{1}{\sqrt{2}}\right) \sqrt{2} \\ &= 1 + \sqrt{2} - 1 = \sqrt{2}. \end{aligned}$$

Thus the diagonal action attains the proposed value and the axial alternatives are more expensive.

At 10, the deterministic Z action reaches 11 in one step, so $V(10) = 1$. Similarly, X sends 01 to 11 in one step. At the goal, the process terminates with value zero. The other goal columns follow by translation symmetry. This explicitly verifies the stochastic-shortest- path Bellman equations whose solution is (15).

7 The complete reinforcement-learning specification

The phrase "one-step cost" can obscure several choices that must be made to define a reinforcement-learning problem. This section spells out the entire model: what a state means operationally, what counts as one time step, what outcomes are observed, when an episode ends, what reward is paid, and whether future rewards are discounted.

7.1 Decision epochs and operational states

A *decision epoch* occurs whenever the agent chooses one quantum instrument. One use of the X channel, one use of the Z channel, one attempt of the random-Y instrument, and one use of the SIC instrument each count as exactly one step. A step is therefore an intervention, not a successful displacement and not an elapsed laboratory time chosen after the outcome is known.

After at least one SIC use, the state supplied to the controller is the most recent SIC token

$$\mathcal{S} = \{00, 10, 01, 11\}.$$

This is not a hidden ontic label inferred from the premeasurement state. By (13), observing token x prepares Π_x, so x is a sufficient classical record of the known *postmeasurement* state. The bit strings are merely convenient names for four observable tokens. The permutations induced by the movement buttons can be learned from anchor–action–SIC experiments as described in Section 5.3.

For a requested goal token g, append one absorbing terminal state

$$\dagger_g = \text{the SIC instrument has just reported } g.$$

Notice the distinction between $x = g$ and $\dagger_g$. At $x = g$, the last SIC outcome was g, but the new task has not yet been certified by a fresh goal outcome. The value at $x = g$ is consequently nonzero. This stronger goal avoids declaring success merely because a theorist says that the system occupies a named state.

7.2 The four actions and all transition probabilities

The action set is

$$\mathcal{A} = \{X, Z, \widetilde{Y}, \mathsf{S}\},$$

where $\widetilde{Y}$ denotes the two-outcome retry instrument and S the four-outcome SIC instrument. Writing $\oplus$ for bitwise addition modulo two and $p = 1/\sqrt{2}$, their controlled transitions are:

action	observed outcome	next operational state and probability
X	one deterministic acknowledgement	$x \mapsto x \oplus 10$ with probability 1
Z	one deterministic acknowledgement	$x \mapsto x \oplus 01$ with probability 1
$\widetilde{Y}$	success s or failure f	$x \mapsto x \oplus 11$ with probability $p = 1/\sqrt{2}$; $x \mapsto x$ with probability $1 - p$
S	token $o \in \mathcal{S}$	$x \mapsto \dagger_g$ if $o = g$; otherwise $x \mapsto o$, with $P(o \mid x) = 1/2$ for $o = x$ and $1/6$ for $o \neq x$

The deterministic acknowledgement of X or Z is not itself a location measurement. Rather, the controller propagates its last measured anchor through an empirically calibrated response permutation. The random-Y outcome says whether that calibrated permutation occurred. The SIC action is the action that creates a new directly observed anchor. Thus the model uses both controlled prediction and occasional measurement, much as a robot uses odometry between landmark observations. It assumes that the learned instrument laws remain stable during an episode; without such stationarity, no controller can propagate any predictive state.

7.3 Reward, cost, stopping time, and the absence of shaping

Let

$$\tau_g = \min\{n \geq 1 : \text{the } n\text{-th chosen instrument is } \mathsf{S} \text{ and reports } g\}$$

be the number of interventions up to and including the successful SIC use. Every chosen action has immediate cost

$$c_t = 1. \tag{16}$$

Equivalently, every action has reward $r_t = -1$. There is no extra reward for applying the "correct" Pauli, no reward for decreasing a supplied coordinate distance, and no terminal bonus. The only special event is that the successful SIC branch ends the episode and therefore eliminates all future costs. With no discounting, the return is

$$G_0 = \sum_{t=0}^{\tau_g - 1} r_t = -\tau_g. \tag{17}$$

Maximizing expected return is exactly the same problem as minimizing the expected number of physical instrument uses:

$$V_g(x) = \inf_{\pi} \mathbb{E}_x^{\pi}[\tau_g]. \tag{18}$$

The unit cost is a modeling choice, albeit a transparent one. If laboratory time, energy, or information acquisition differs by instrument, replace $c_t = 1$ by action-dependent costs $c_a > 0$. The Bellman proof and the resulting geometry must then be recomputed. In particular, calling the SIC measurement "free" would change the geometry.

7.4 The undiscounted Bellman equation

The hodological construction uses the stochastic-shortest-path convention

$$\boxed{\gamma = 1.} \tag{19}$$

For any nonterminal anchor x, define

$$Q_g(x, a) = 1 + \sum_{y \in \mathcal{S}} P_a(y \mid x) V_g(y). \tag{20}$$

The terminal transition has no continuation term. The optimality equation is

$$\boxed{V_g(x) = \min_{a \in \mathcal{A}} Q_g(x, a).} \tag{21}$$

Using $\gamma = 1$ is legitimate here even though many introductory RL algorithms emphasize $\gamma < 1$. This is an episodic proper stochastic-shortest-path problem: the policy "repeat the SIC" terminates with positive probability on each attempt, all costs are positive, and the optimal expected stopping time is finite. Discounting is therefore not needed to make the objective well defined. More importantly, expected hitting time is the quantity being interpreted as spatial distance.

8 Full exact action analysis at $\gamma = 1$

Choose $g = 00$; all other goals follow by symmetry. There are three distance shells: 0 for 00, 1 for $10, 01$, and 2 for 11. Write

$$c = \frac{8 + \sqrt{2}}{3}, \qquad v_0 = c, \qquad v_1 = c + 1, \qquad v_2 = c + \sqrt{2}.$$

The following table evaluates *every* action, not only the optimal one. Within the edge row, "toward" is X from 10 and Z from 01; "away" is the other axial action.

current shell	action	exact Q-value	optimal?
self	X or Z	$c + 2$	no
self	$\widetilde{Y}$	$c + 2$	no
self	SIC	c	yes
edge	axial toward goal	$c + 1$	yes
edge	axial away from goal	$c + 1 + \sqrt{2}$	no
edge	$\widetilde{Y}$	$c + 2$	no
edge	SIC	$c + 1 + (2 + \sqrt{2})/9$	no
diagonal	X or Z	$c + 2$	no
diagonal	$\widetilde{Y}$	$c + \sqrt{2}$	yes
diagonal	SIC	$c + \sqrt{2} + (8 - 5\sqrt{2})/9$	no

Here are the calculations behind the table. At the self anchor, either axial action reaches an edge, so $Q_X = Q_Z = 1 + v_1 = c + 2$. Random Y fails at self with probability $1 - p$ and succeeds to the diagonal with probability p, whence

$$Q_{\widetilde{Y}} = 1 + (1-p)v_0 + pv_2 = 1 + c + p\sqrt{2} = c + 2.$$

The SIC terminates with probability 1/2, sends the system to either edge with total probability 1/3, and to the diagonal with probability 1/6:

$$Q_{\mathsf{S}} = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2 = c.$$

At an edge, the correct axial action incurs one step and reaches self, giving $1 + v_0 = v_1$. The other axial action reaches the diagonal. Random Y either stays at the same edge or reaches the other edge, so its continuation value is always v_1. A SIC use terminates with probability 1/6, has an edge continuation with total probability 2/3, and a diagonal continuation with probability 1/6. Therefore

$$\begin{aligned} Q_{\mathsf{S}}(\text{edge}) &= 1 + \frac{2}{3}v_1 + \frac{1}{6}v_2 \\ &= v_1 + \frac{2+\sqrt{2}}{9} > v_1. \end{aligned}$$

At the diagonal, either axial action reaches an edge and costs $1 + v_1 = c + 2$. Random Y succeeds to self with probability p and otherwise remains at the diagonal:

$$\begin{aligned} Q_{\widetilde{Y}}(\text{diagonal}) &= 1 + (1-p)v_2 + pv_0 \\ &= v_2, \end{aligned}$$

where the last equality is exactly the geometric-retry identity $p(v_2 - v_0) = p\sqrt{2} = 1$. Finally,

$$\begin{aligned} Q_{\mathsf{S}}(\text{diagonal}) &= 1 + \frac{1}{3}v_1 + \frac{1}{2}v_2 \\ &= v_2 + \frac{8-5\sqrt{2}}{9} > v_2. \end{aligned}$$

Both strict gaps are positive. The complete optimal policy is consequently

$$\pi_g^*(x) = \begin{cases} \mathsf{S}, & x = g, \\ \text{the axial action taking } x \text{ to } g, & d(x,g) = 1, \\ \widetilde{Y}, & d(x,g) = \sqrt{2}. \end{cases} \tag{22}$$

Numerically, $c \simeq 3.138071$. The self, edge, and diagonal values are, respectively, 3.138071, 4.138071, and 4.552285. Subtracting the unavoidable goal-reporting cost v_0 gives $0, 1, \sqrt{2}$, the desired square radii.

9 What discounting would change

For $0 \leq \gamma < 1$, the discounted step-cost objective would instead be

$$V_g^{(\gamma)}(x) = \inf_\pi \mathbb{E}_x^\pi \left[\sum_{t=0}^{\tau_g - 1} \gamma^t \right] = \inf_\pi \frac{1 - \mathbb{E}_x^\pi[\gamma^{\tau_g}]}{1-\gamma}. \tag{23}$$

It has Bellman equation

$$V_g^{(\gamma)}(x) = \min_a \left\{ 1 + \gamma \sum_{y \in \mathcal{S}} P_a(y \mid x) V_g^{(\gamma)}(y) \right\}. \tag{24}$$

For a deterministic stopping time, (23) is a monotone transform of the number of steps. For a random stopping time it depends on the entire distribution of τ_g, not only its mean. It preferentially weights chances of very early completion. A common alternative reward, zero on ordinary transitions and $+1$ on the terminal SIC outcome, maximizes $\mathbb{E}[\gamma^{\tau_g - 1}]$. For $0 < \gamma < 1$ this selects the same policies as minimizing (23), but neither objective is the expected hitting time that defines the present hodological metric.

The distinction can be solved exactly in this four-state example. Again let v_0, v_1, v_2 denote the self, edge, and diagonal discounted costs. There are three policy regimes. Define

$$\gamma_* = \frac{3 + 3\sqrt{2} - \sqrt{15 + 12\sqrt{2}}}{2} \simeq 0.7941944732. \tag{25}$$

At $\gamma = 0$, every action has the same one-step discounted cost, so every first action ties. For $0 < \gamma \le 1/2$, a SIC use is optimal in all three shells. The following exact solution also extends continuously to $\gamma = 0$:

$$v_0 = \frac{6 - 2\gamma}{6 - 5\gamma}, \qquad v_1 = v_2 = \frac{6}{6 - 5\gamma}. \tag{26}$$

The equality $v_1 = v_2$ shows that strong discounting erases the distinction between an edge and the diagonal. To derive these expressions, observe that from any non-goal anchor a SIC reports g with probability $1/6$; under the symmetric all-SIC solution its total nonterminal continuation is $(5/6)v_1$. At self, the nonterminal probability is $1/2$.

For $1/2 \le \gamma \le \gamma_*$, SIC is optimal at self, an axial move is optimal at an edge, and SIC remains optimal at the diagonal. With

$$\Delta = 2\gamma^3 - 6\gamma^2 - 9\gamma + 18,$$

the exact values are

$$v_0 = \frac{2(9 - \gamma^2)}{\Delta}, \qquad v_1 = \frac{3(-2\gamma^2 + 3\gamma + 6)}{\Delta}, \qquad v_2 = \frac{6(\gamma + 3)}{\Delta}. \tag{27}$$

They follow by solving the three linear policy-evaluation equations

$$v_0 = 1 + \gamma\left(\frac{1}{3}v_1 + \frac{1}{6}v_2\right), \quad v_1 = 1 + \gamma v_0, \quad v_2 = 1 + \gamma\left(\frac{1}{3}v_1 + \frac{1}{2}v_2\right).$$

For $\gamma_* \le \gamma \le 1$, the undiscounted policy structure is restored: SIC at self, axial movement at an edge, and retry-Y at the diagonal. Put

$$a_\gamma = 1 - \gamma\left(1 - \frac{1}{\sqrt{2}}\right).$$

Then an economical exact form is

$$v_0 = \frac{1 + \gamma/3 + \gamma/(6a_\gamma)}{1 - \gamma^2/3 - \gamma^2/(6\sqrt{2}a_\gamma)}, \tag{28}$$

$$v_1 = 1 + \gamma v_0, \qquad v_2 = \frac{1 + \gamma v_0/\sqrt{2}}{a_\gamma}. \tag{29}$$

The first transition occurs when the edge is indifferent between reporting and moving, which reduces exactly to $\gamma = 1/2$. The second occurs when the diagonal is indifferent between reporting and retrying Y. Substituting (27) gives

$$\gamma^2 - 3(1+\sqrt{2})\gamma + 3 + \frac{3\sqrt{2}}{2} = 0,$$

whose only root in $[0, 1]$ is (25). Direct substitution checks that all unselected actions have no smaller Q-value in their stated intervals. At a boundary the adjacent policies tie.

At $\gamma = 1$, (28)– (29) reduce to

$$(v_0, v_1, v_2) = (c, c+1, c+\sqrt{2}).$$

Thus the Euclidean square is not invariant under arbitrary discounting. It is specifically a geometry of expected undiscounted intervention count. The computed phase diagram in Fig. 1 shows the approach to that limit.

10 A compact derivation of the SIC reporting baseline

For reference, we can derive the common baseline in a compact three-shell form. As specified in Section 7, goal g means: perform the SIC action and observe outcome token g. Every movement and every SIC report costs one intervention. If a SIC outcome is not g, the episode continues from the newly prepared state Π_o. After the first SIC outcome this is a fully observed finite MDP, because the most recent token is a sufficient operational state.

By square symmetry, write v_0, v_1, v_2 for optimal cost when the current anchor is respectively the goal, an axial neighbor, or the opposite corner. The proposed policy reports at the goal, moves deterministically from an edge to the goal, and retries the diagonal displacement from the opposite corner. The two movement rules give

$$v_1 = 1 + v_0, \qquad v_2 = \sqrt{2} + v_0. \tag{30}$$

At the goal state, the SIC reports g with probability $1/2$, terminating the episode. Its two edge outcomes have total probability $1/3$, and its diagonal outcome has probability $1/6$. Consequently

$$v_0 = 1 + \frac{1}{3}v_1 + \frac{1}{6}v_2 \tag{31}$$
$$= 1 + \frac{1}{3}(v_0+1) + \frac{1}{6}(v_0+\sqrt{2}), \tag{32}$$

so

$$\boxed{v_0 = c := \frac{8+\sqrt{2}}{3}}. \tag{33}$$

It follows that the full reported-goal value matrix is

$$\boxed{V_g(x) = c + D(x, g).} \tag{34}$$

The remaining Bellman inequalities confirm optimality. At the goal, moving first costs $c+2 > c$. At an edge, reporting immediately costs

$$1 + \frac{2}{3}v_1 + \frac{1}{6}v_2 > v_1,$$

while the axial move attains v_1. At the diagonal, reporting costs

$$1 + \frac{1}{3}v_1 + \frac{1}{2}v_2 > v_2,$$

and an axial two-step route costs $c+2 > c+\sqrt{2}$; diagonal retry attains v_2. Thus (34) is the optimal Bellman solution, not merely the value of a selected policy.

The nonzero self-value $V_g(g) = c$ is the common overhead of requesting a stochastic SIC report. Hodological separation is obtained by subtracting this goal-specific self baseline:

$$V_g(x) - V_g(g) = D(x, g). \tag{35}$$

The exact Euclidean square therefore survives when places and goals are both defined entirely by experienced SIC outcomes.

Before the first SIC action, an agent with an unknown preparation occupies an epistemic initial condition rather than one of four known anchors. Its first SIC outcome both reports a token and prepares the associated state. Subsequent tracking uses only that token, chosen actions, and observed success/failure events; the policy needs no concealed binary-coordinate variable.

11 Exact computation and finite-sample simulation

The companion program `sic_anchored_square.py` constructs all density matrices and Kraus operators, solves the four stochastic-shortest-path problems, runs Monte Carlo navigation, and learns randomly renamed Pauli permutations from anchor–action–SIC experiments. Eight focused tests verify Kraus completeness, branch preparation, the $1/2, 1/6$ response kernel, the analytic Bellman solution, every undiscounted action value, the three discounted policy regimes, and optimal-policy structure.

At seed 20260813, exact value iteration gives

$$\max_{x,g} |V_g(x) - c - D(x, g)| = 2.58 \times 10^{-14},$$

and baseline subtraction recovers D to 9.10×10^{-15}. Across 50,000 Monte Carlo episodes for each of the 16 ordered pairs, maximum mean-cost error is 0.02831. In 500 independent opaque-label trials per sample size, the complete four-action response permutations are recovered in every trial at 100 samples per anchor/action pair. Across a 503-point discount scan, the closed forms in Section 9 agree with numerical Bellman iteration to 3.02×10^{-14}.

12 Why D is exactly a two-dimensional Euclidean metric

The entries of D already match the familiar side and diagonal lengths of a unit square. Schoenberg's test certifies this without presupposing the coordinates in (9).

First square every entry:

$$D^{\circ 2} = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}. \tag{36}$$

Every row mean, every column mean, and the grand mean of this matrix equal one. With

$$J = I - \frac{1}{4}\mathbf{1}\mathbf{1}^T,$$

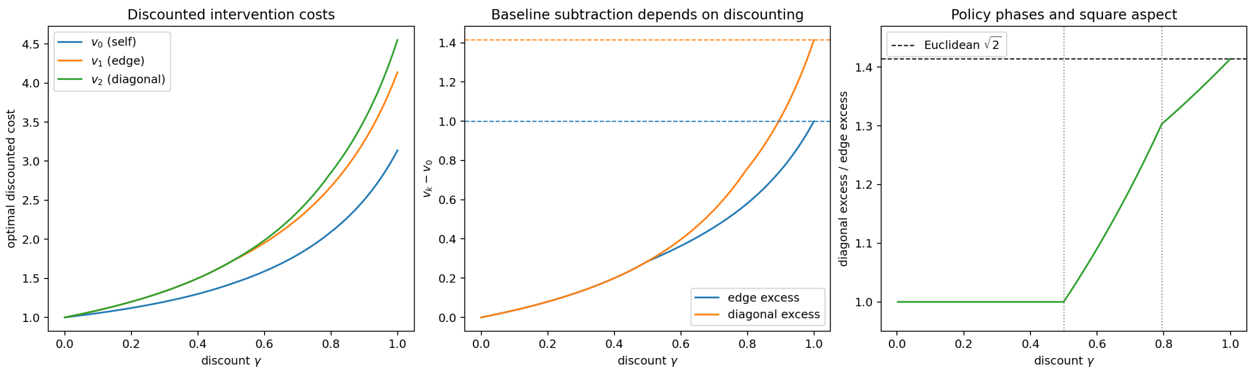

Figure 1: **Discounting changes both value and geometry.** Left: exact discounted self, edge, and diagonal intervention costs. Center: after subtracting the self reporting baseline, the edge and diagonal separations reach 1 and $\sqrt{2}$ only at $\gamma = 1$. Right: the diagonal-to-edge ratio and the two exact policy thresholds $\gamma = 1/2$ and $\gamma = \gamma_*$. Strong discounting makes all non-goal SIC-report policies optimal and collapses edge and diagonal into one shell.

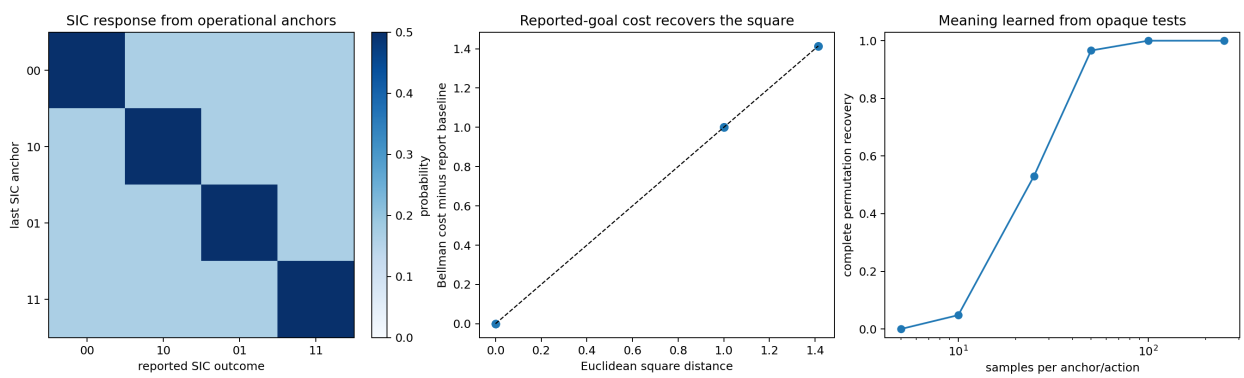

Figure 2: **The square grounded in SIC outcomes.** Left: after an observed SIC anchor, the same outcome has probability 1/2, every other outcome has probability 1/6, and every branch prepares the state named by its token. Center: subtracting the common reporting baseline from exact Bellman costs recovers side 1 and diagonal $\sqrt{2}$. Right: randomly renamed Pauli response permutations are learned from future SIC tests without exposing binary coordinates.

the double-centered Gram matrix is

$$B = -\frac{1}{2} J D^{\circ 2} J \tag{37}$$

$$= \begin{pmatrix} \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}. \tag{38}$$

To see the geometry directly, center the square coordinates:

$$X = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{39}$$

Multiplication gives $XX^T = B$. Therefore B is positive semidefinite. The two columns of X are orthonormal, so the eigenvalues of B are

$$1,\ 1,\ 0,\ 0.$$

It has rank two, and classical multidimensional scaling reconstructs the four rows of X, up to rotation and reflection. This proves intrinsically that the hodological metric (15) is an exact two-dimensional Euclidean metric.

13 What the theorem proves—and what it assumes

The theorem proves three useful facts.

1. The Kraus operators form valid quantum instruments.
2. The desired metric is achieved by a concrete policy.
3. The triangle inequality is a global certificate against every adaptive shortcut.

It does not derive the metric from an otherwise neutral quantum dynamics. The desired length is inserted into the success law $p_h = 1/d_G(\mathsf{e}, h)$. In the square, the Euclidean diagonal is obtained because the diagonal action was assigned success probability $1/\sqrt{2}$. A different probability would produce a different hodological diagonal.

It also does not imply that instrument outcomes localize the agent. In a random-unitary retry action, success/failure reports whether a chosen translation occurred; it need not reveal the starting state. Finally, the four qubit states form a tetrahedron in Bloch-state geometry even though their optimal costs form a square. This separation is intentional: hodological distance measures difficulty of controlled goal achievement, not quantum state overlap.

The SIC extension in Sections 5.3 and 10 repairs the location-token problem in a precise sense: an outcome prepares and therefore identifies the *postmeasurement* anchor, and place goals are future SIC outcomes. It does not retrospectively identify the unknown premeasurement state, nor make the retry movement outcome itself location-informative. Its virtue is that the agent's current-state token and its goal tokens now arise from one actual measurement alphabet.

14 A compact checklist for using the theorem

Before applying the retry theorem to a new construction, check:

1. Is every desired displacement represented by a valid group element?
2. Is the orbit faithful after quotienting global phase and the fiducial stabilizer?
3. Does the side of multiplication match the side of metric invariance?
4. Has the metric been normalized so every $p_h \leq 1$?
5. Does failure truly leave the operational coordinate unchanged?
6. Is the goal defined by the physical orbit, a predictive equivalence class, or an external group-product monitor? These are not interchangeable.
7. Is $p_h = 1/d_G(\mathsf{e}, h)$ physically motivated, learned, or simply supplied by design?

For Eq. (43), all algebraic conditions hold exactly. The remaining scientific limitation is the last one: the Euclidean success law is engineered rather than derived.